



LTR9200-T18

10-Bit SAR A/D Converter

Preliminary Technical Data

DESCRIPTION

The LTR9200-T18 is a 10-bit, low power, high precision Successive Approximation Register (SAR) Analog-to-Digital Converter (ADC) Virtual Component (IP). It is designed to meet power saving requirements of portable battery powered systems. This design goal is achieved by providing to the SoC designers a complete low-power A/D converter solution with multiplexed analog input channels.

The LTR9200 is designed to achieve high speed data conversion with ± 1 LSB Differential Nonlinearity (DNL) and Integral Nonlinearity (INL) with very low power consumption. Operates from a 3.3V power supply for analog and 1.8V for digital, the LTR9200 only consumes 2mA current in active mode and 100nA in shutdown mode. These features make the LTR9200 an ideal choice for low power high speed applications. The LTR9200 is configurable to accept 8 single-ended inputs or 4 fully differential inputs and digitizes the input at a maximum 1 MSPS throughput rate. It features build-in sample and hold circuitry with offset calibration.

This IP block is designed in TSMC 0.18 micron CMOS technology and can be retargeted to other foundry process technologies.

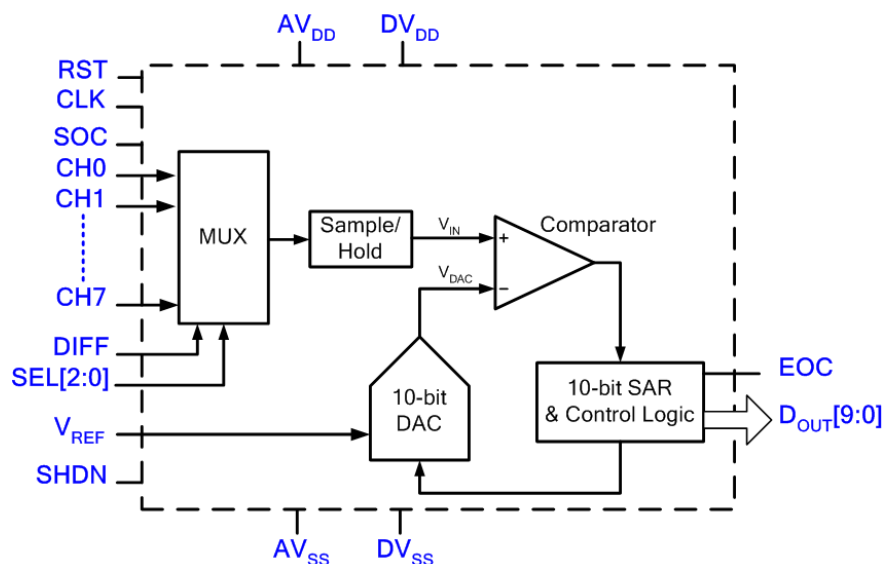
FEATURES

- Fast throughput rate: 1 MSPS
- 10-bit resolution
- ± 1 LSB max DNL
- ± 1 LSB max INL
- Configurable analog inputs:
 - 8-channel single-ended inputs
 - 4-channel fully differential inputs
- Analog input voltage range: 0 to 3.3V
- Power supply: 3.3V analog and 1.8V digital
- Low power technology:
 - 2mA typical operating current
 - 100nA typical shutdown current
- Build-in sample and hold circuitry with offset calibration
- Industrial temp range: -40°C to $+85^{\circ}\text{C}$

TYPICAL APPLICATIONS

- Sensor interface/Data acquisition
- Automotive sensors
- Multi-sensor wireless network
- Building/home control
- High-end mobile phones
- Biomedical devices/Industrial automation
- Battery powered/Handheld systems

FUNCTIONAL BLOCK DIAGRAM



PIN DESCRIPTION

PIN NAME	DESCRIPTION
CLK	Conversion Clock Input
RST	Reset Input
SHDN	Shutdown Input
DIFF	Differential Mode Select Input. Determine if the conversion will be done using single-ended or fully differential input configuration mode.
V _{REF}	Voltage Reference Input. The voltage applied to this input determines the analog input voltage range.
CH[0-7]	Analog Input Channels 0 to 7. Each pair of channels can be configured to be used as two independent channels in single-ended mode or as a fully differential input where one channel is V _{IN+} and one channel is V _{IN-} .
SEL[0-2]	Analog Input Channel Select Input
SOC	Start of Conversion Input. This logic input initiates conversions on the LTR9205-T25.
EOC	End of Conversion Output. EOC is asserted at the end of each conversion cycle.
D _{OUT} [0-9]	Data Bits. Parallel digital outputs that provide the conversion result. The data is valid when EOC is asserted.
AV _{DD}	Analog Power Supply Input
DV _{DD}	Digital Power Supply Input
AV _{SS}	Analog Ground
DV _{SS}	Digital Ground

INPUT SELECTION TABLE

SEL[0-2]	DIFF	VIN+	VIN-
000	0	ch0	internal
001	0	ch1	internal
010	0	ch2	internal
011	0	ch3	internal
100	0	ch4	internal
101	0	ch5	internal
110	0	ch6	internal
111	0	ch7	internal
X00	1	ch0	ch1
X01	1	ch2	ch3
X10	1	ch4	ch5
X11	1	ch6	ch7

ELECTRICAL SPECIFICATIONS

The specifications apply over the full operating temperature range: -40°C to $+85^{\circ}\text{C}$. Typical values are at ambient temperature $T_A = 25^{\circ}\text{C}$. Test conditions: $AV_{DD} = 3.3\text{V}$, $DV_{DD} = 1.8\text{V}$, $V_{REF} = 3.0\text{V}$, $f_{CLK} = 16\text{ MHz}$, $f_{CONV} = 1.0\text{ Msps}$

Parameters	Symbol	MIN	TYP	MAX	Units	Conditions
Analog Supply Voltage	AV_{DD}	3.0	3.3	3.6	V	
Digital Supply Voltage	DV_{DD}	1.6	1.8	2	V	
Operating Current	I_{DD}		2		mA	
Resolution	N		10		Bit	
Analog Reference Voltage	V_{REF}	0		3.3	V	
Analog Input Voltage	V_{IN}	0		3.3	V	
Eqv. Input Capacitance	R_{IN}		1		pF	
Clock frequency	f_{CLK}	14	16	18	MHz	
Sample Time	t_{sample}	2			clk cycles	
Conversion Time (Note1)	t_{conv}			16	clk cycles	
Conversion Rate	f_{conv}	0.875	1		Mbps	
Integral Non-Linearity	INL	-1		1	LSB	-40°C to $+85^{\circ}\text{C}$
Diff Non-Linearity	DNL	-1		1	LSB	-40°C to $+85^{\circ}\text{C}$
Offset	OFF	-1		1	LSB	
Gain	GAIN	-1		1	LSB	
Shutdown Current	I_{SHDN}		100		nA	

Note 1: This conversion time includes the sample time.

FUNCTIONAL DESCRIPTION

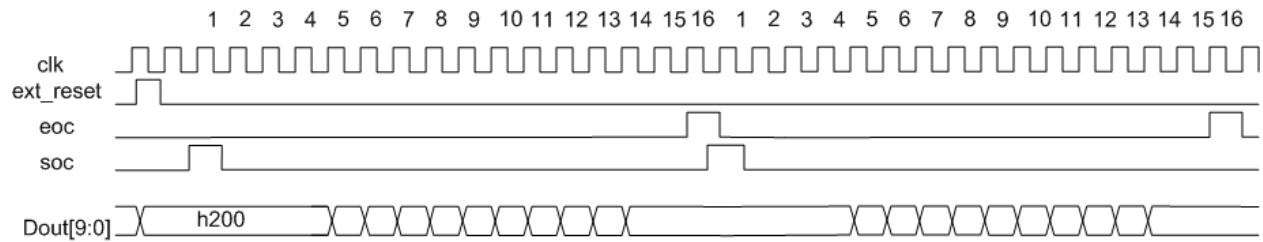


Figure 1 - LTR9200-T18 Timing Diagram

REVISION HISTORY:

Version No.	Issue Date	Description
1.0	Mar 14, 2005	First release, Preliminary Technical Data.

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